

CHARLSEY R. TOMASSETTI

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Department of Physics, Applied Physics, and Astronomy

Binghamton University-SUNY, Binghamton, New York 13902, USA

EDUCATION

Binghamton University, State University of New York

Doctor of Philosophy in Physics

Expected May 2026

Advisor: Elena R. Margine

Co-supervisor: Alexey Kolmogorov

Bachelor of Science in Physics

May 2020

Bachelor of Arts in English

May 2020

EXPERIENCE

Educational Opportunity Program's Summer Bridge Program

July 2025 - August 2025

Math Tutor

Binghamton, NY

- Tutored incoming college freshmen (30+) in algebra and pre-algebra individually and in small groups
- Developed and implemented tutorial plans tailored to each student's academic needs
- Improved student's math comprehension and work ethic

Physics Outreach Project

September 2023 - May 2024

Volunteer

Binghamton, NY

- Delivered engaging and educational science demonstrations to groups of 10+ elementary students (K-5) for the purpose of fostering and inspiring interest in science

Binghamton University, State University of New York

August 2020 - May 2022

Teaching Assistant

Binghamton, NY

- Mentored, supervised and lectured groups of 80+ students in understanding general physics theory, through guided group activity and one-on-one tutoring
- Developed and administered tests and quizzes in a timely manner
- Constructed own problem sets and devised challenging test questions

American Physical Society New York State Section

November 2019 - December 2019

Newsletter Editor

Binghamton, NY

- Collected and synthesized research and presentations from two previous APS meetings for yearly newsletter
- Presented recent outreach efforts by the New York State Section in clean, easily understandable prose

PUBLICATIONS

G. P. Kafle, C. R. Tomassetti, I. I. Mazin, A. N. Kolmogorov, and E. R. Margine, *Ab initio* study of Li-Mg-B superconductors, *Physical Review Materials* **6**, 084801 (2022).

S. Kharabadze, M. Meyers, C. R. Tomassetti, E. R. Margine, I. I. Mazin, and A. N. Kolmogorov, Thermodynamic stability of Li-B-C compounds from first principles, *Physical Chemistry Chemical Physics* **25**, 7344 (2023).

C. R. Tomassetti, G. P. Kafle, E. T. Marcial, E. R. Margine, and A. N. Kolmogorov, Prospect of high-temperature superconductivity in layered metal borocarbides, *Journal of Materials Chemistry C* **12**, 4870 (2024).

C. R. Tomassetti, D. Gochitashvili, C. Renskers, E. R. Margine, and A. N. Kolmogorov, First-principles design of ambient-pressure $\text{Mg}_x\text{B}_2\text{C}_2$ and Na_xBC superconductors, *Physical Review Materials* **8**, 114801 (2024).

D. Gochitashvili, C. R. Tomassetti, E. R. Margine, and A. N. Kolmogorov, High- T_c Ag_xBC and Cu_xBC superconductors accessible via topochemical reactions. *Journal of Materials Chemistry C*, **13(36)**, 18924–18935 (2025).

CONFERENCE PRESENTATIONS

Oral Presentations

- C. R. Tomassetti, G. P. Kafle, I. I. Mazin, A. N. Kolmogorov, and E. R. Margine. (2023, March). *Stability of layered Li-Mg-B compounds from first-principles*. Minisymposium on computational design, understanding and discovery of novel materials, March Meeting 2023 of the American Physical Society, Las Vegas, Nevada.
- C. R. Tomassetti, G. P. Kafle, E. Marcial, I. I. Mazin, E. R. Margine, and A. N. Kolmogorov. (2024, March). *Understanding and predicting high- T_c superconductivity in metal borocarbides*. Minisymposium on computational design, understanding and discovery of novel materials, March Meeting 2024 of the American Physical Society, Minneapolis, Minnesota.
- C. R. Tomassetti, D. Gochitashvili, C. Renskers, E. R. Margine, and A. N. Kolmogorov. *First-principles design of ambient-pressure $\text{Mg}_x\text{B}_2\text{C}_2$ and Na_xBC superconductors*. Minisymposium on superconductors, Global Physics Summit 2025 of the American Physical Society, Anaheim, California.

Poster Presentations

- C. R. Tomassetti, G. P. Kafle, I. I. Mazin, A. N. Kolmogorov, and E. R. Margine. (2022, June). *Ab initio* study of Li-Mg-B superconductors. Poster session presented at the 2022 School on Electron-Phonon Physics from First Principles, Austin, Texas.
- C. R. Tomassetti, G. P. Kafle, E. Marcial, I. I. Mazin, E. R. Margine, and A. N. Kolmogorov. (2023, October). Understanding and predicting high- T_c superconductivity in metal borocarbides. Poster session presented at the Fall 2023 Meeting of the New York State Section of the American Physical Society, Rochester, New York.
- C. R. Tomassetti, G. P. Kafle, E. Marcial, I. I. Mazin, E. R. Margine, and A. N. Kolmogorov. (2024, April). Understanding and predicting high- T_c superconductivity in metal borocarbides. Poster session presented at the Materials Research Society Binghamton Student Chapter Smart Energy Symposium 2024, Binghamton, New York.

GRANTS AND SPONSORSHIPS

2022 School on Electron-Phonon Physics from First Principles, Travel Support Grant	2022
Binghamton University, Graduate School Travel Grant	2022
American Physical Society, DCOMP Travel Award	2024
Binghamton University, Graduate School Travel Grant	2025

TECHNICAL STRENGTHS

**Materials Modeling Softwares
Tools**

Quantum ESPRESSO, EPW, VASP
Vim, Emacs